

Sanath Kumar Das

Chattogram, Bangladesh

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LinkedIn — Google Scholar — Faculty Profile

Research Interests

- Physiological Signal Decoding using Deep Representation Learning
- Brain–Computer Interfaces and Neuroprosthetic Systems
- Intelligent Control and Multi-Agent Robotic Systems
- Rehabilitation Engineering

Education

Bangladesh University of Engineering and Technology (BUET)

Master’s Degree in Electrical and Electronic Engineering (Ongoing)
CGPA: 3.66 / 4.00

Bachelor’s Degree in Electrical and Electronic Engineering 2018–2023
CGPA: 3.51 / 4.00
Major: Communication and Signal Processing

Publications

- **Sanath Kumar Das**, Arpan Sur, Md. Farhad Hossain. Collaborative Path Planning and Target Search in Multi-Robot Systems with PID-Controlled Uniform Area Partitioning. *2024 International Conference on Computer and Information Technology (ICCIT)*. DOI: 10.1109/ICCIT64611.2024.11022590
- **Sanath Kumar Das**, Sudipta Saha, Fatin Fardaous, et al. Aquatic Robot for Trash Removal and Water Quality Monitoring Built from Recycled Materials. *2024 13th International Conference on Electrical and Computer Engineering (ICECE)*. DOI: 10.1109/ICECE64886.2024.11024687

Research Experience

Master’s Thesis (Ongoing)

Representation Learning for Imagined Word Decoding from EEG Signals

- Processing multi-channel EEG signals and investigating time-frequency, spatial, and connectivity-based feature extraction methods for imagined word decoding.
- Implementing deep learning architectures — including CNNs, Transformers, hybrid networks, and graph neural networks — to model spatial–temporal neural dynamics from EEG signals.
- Evaluating handcrafted and learned representations with emphasis on robustness, generalization, and neurophysiological interpretability for real-world BCI applications.

Bachelor’s Thesis

A Collaborative Multi-Robot Coverage Path Planning and Efficient Target Search System

- Designed cooperative multi-robot exploration system for unknown environments.
- Implemented coverage path planning and efficient target search algorithms.
- Simulation and timing analysis performed using CoppeliaSim and Python.

Teaching Experience

Lecturer, Department of Electrical and Electronic Engineering

East Delta University, Chattogram

May 2024 – Present

- Prepare course materials including lecture notes, assignments and assessments plans.
- Deliver lectures and mentor undergraduate students for academic projects.

Technical Skills

Programming: Python, C, C++, MATLAB, Verilog, System Verilog

Machine Learning / Data Science: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib

BCI Tools: MNE-Python, Pyriemann

Simulation / Tools: Simulink, Proteus, Quartus, CoppeliaSim

Academic Projects

Selected laboratory and course projects completed during undergraduate studies

- Design and hardware implementation of PPG-Based Pulse Monitoring and Counting System Using an IR sensor and amplifier
- Aquatic Multi-Purpose Robot for Water Surface Trash Cleaning and Water Quality Monitoring
- Simulation of 8-bit TDM-PCM Transmitter and Receiver System in Proteus
- Real-Time Human Detection and Social Distance Estimation Using YOLOv3
- IoT-Enabled Fire and Gas Leak Detection and Automatic Prevention System Using ESP-32 Microcontroller

Awards and Achievements

- RISE Research Grant 2022 – Received 30,000 BDT research grant.
- Scholarship in secondary and higher secondary examinations.

Leadership and Volunteer Activities

- General Secretary, BUET Robotics Society (2022–2023)
- Instructor at Basic Robotics Workshop organized by BUET Robotics Club

References

Dr. Shaikh Anowarul Fattah Professor, Department of EEE, BUET

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Dr. Md. Farhad Hossain Professor, Department of EEE, BUET

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